Road Safety & Congestion KPI Reference

Definitions and thresholds for interpreting traffic conditions · Knowledge base v1.0

1. Congestion metrics

Metric	Definition
Travel Time Index (TTI)	Observed travel time ÷ free-flow travel time on the same segment. 1.0–1.3 free/light, 1.3–1.8 moderate, 1.8–2.5 heavy, above 2.5 severe.
Delay (min)	Observed minus free-flow travel time for the route in question — the primary number to quote to operations.
Peak spreading	Duration for which a corridor's TTI stays above 1.8 in a day; growth over weeks signals structural capacity issues.

2. Incident severity classes

Class	Examples	Operational response
Critical	Multi-vehicle accident, full closure of a major corridor	Immediate re-route advice; notify operations duty officer.
Major	Lane-blocking accident, emergency roadworks	Advise alternates; monitor until cleared.
Moderate	Shoulder incident, planned roadworks with lane restriction	Note in the corridor report.
Minor	Broken-down vehicle off-carriageway, minor flooding cleared	Log only.

3. Safety-relevant patterns

- Recurring incidents at the same interchange across multiple weeks — flag for engineering review.
- Congestion shockwaves upstream of merges during peaks — candidates for ramp-metering studies.
- High closure frequency on a corridor combined with rising TTI — prioritize for the resilience program.

4. Reporting thresholds

Escalate in the daily brief when: any corridor sustains TTI above 2.5 for 30+ minutes; a critical incident affects a strategic route; or planned works overlap on parallel corridors, removing the natural alternate.